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The Program in Applied and Computational Mathematics
Princeton University, Fine Hall
Princeton, NJ 08544-1000, USA

Dear Members of the Search Committee,

I am writing to apply to the Ph.D. in Applied and Computational Mathematics, with the intention to register to the Joint Graduate Degree Program in Neuroscience. I am currently studying towards a Master of Advanced Study in Applied Mathematics at the University of Cambridge, and expect to graduate in July 2021. I believe my academic background in mathematics, my research experience in applied mathematics in deep learning and computational neuroscience, as well as my undergraduate teaching experience make me a strong candidate for this collaborative program at Princeton University.

My goal is to apply robust mathematical principles to expand the understanding of learning mechanisms, motivated by questions arising at the intersection of deep learning and neuroscience. This theoretical work requires an interdisciplinary quantitative approach, combining biological research with mathematical analysis in artificial systems. Inherently interdisciplinary and interdepartmental, such approaches are central to The Program in Applied and Computational Mathematics (PACM). My ongoing master's degree in applied mathematics builds upon my existing knowledge and is providing me with the tools needed for further postgraduate study. I am planning to write my master's essay on leveraging neural dynamics for motor control, under the supervision of Professor Stephen Eglen. I am currently taking courses in applied and computational analysis, alongside courses in stochastic processes and theoretical analysis and PDEs. These courses will strengthen the theoretical foundations I acquired through my undergraduate studies in Pure and Applied Mathematics at the Université de Montréal, where I graduated top of the cohort.

¶: High-level:
- My goal,
- why PACM,
- my background

For a year and a half alongside my undergraduate studies, I worked as a scholarship-funded research assistant at Mila — Quebec AI Institute, in affiliation with the Université de Montréal, and under the supervision of Professor Guillaume Lajoie. During this internship, I drew inspiration from cortical networks in order to investigate the network-level computational advantages of various timescales of single-neuron adaptation, within an artificial setting. This provided me with a strong theoretical and programming background in sequential processing with artificial neural net-

works. This research – which is still ongoing – connects computational neuroscience and AI, and led to numerous refereed conference posters and talks, with first-author work awaiting submission for publication. Through research at the intersection of these rapidly expanding fields, I became increasingly aware of a need to lay out theoretical foundations of intelligence, both artificial and biological. PACM at Princeton suits my interests well, as it draws together applied mathematics, neuroscience, and machine learning with its ties with the Princeton Neuroscience Institute and the Center for Statistics and Machine Learning.

As a Ph.D. candidate at PACM, I would be interested in formalizing the role of information and sensory integration in shaping emerging nonlinear dynamics, both within biological and artificial neural networks. For instance, how do single-neuron physiological properties interact with computational sensory processing mechanisms (i.e., perception, attention) to shape the non-autonomous dynamics of neuronal systems? How does this interaction vary depending on the learning timescales considered? Such an approach would utilize tools from the theory of dynamical systems, stochastic modeling, information theory, and control theory. I believe my interests could tie in well with research by Professor Weinan E and potential collaborations with the Princeton Neuroscience Institute, such as with Professor Jonathan Pillow. My academic background and research experience have provided me with the interest, and the tools, to pursue this research.

¶: Specific research interests, possible advisors.

This is my advisor. This letter is how he heard of me.

Sharing my knowledge is fundamental to my academic approach. As I was learning the underlying mechanisms driving modern deep learning research, I became a mentor for the S.E.U.R. university project, presenting numerous conferences on the mathematics of AI in educational institutions across the Montréal. Alongside this, I have been extensively involved in the mathematical community at the Université de Montréal since my first year as an undergraduate. My contributions include co-founding the Mathematical Modeling Club, through which I led workshops on mathematical modeling and organized our participation in an international contest in applied mathematics. I also worked as a teaching assistant (TA) at the Université de Montréal for a first-year course in Real Analysis. It was a privilege to be the only third-year student teaching on this highly challenging course. I am eager to get involved with similar projects at Princeton University and I am confident in my ability to effectively communicate my knowledge and research in applied mathematics.

In the future, I aim to establish nonlinear dynamics as a governing principle of learning, both artificial and biological. This contributes to my academic goal to lead an interdisciplinary research coalition dedicated to laying out the theoretical foundations of intelligence, both during my Ph.D. and after. These goals act as progression towards a professorship at a world-leading research institution, in which I would take an active role in research and teaching. Working at PACM and its related institutes will provide me with opportunities to work in applied mathematics at the forefront of computational neuroscience and deep learning, in addition to cutting-edge analytical and practical methods to deliver high-impact research.

Thank you for your time and consideration; I look forward to hearing from you.

Sincerely,

Victor Geadah